

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

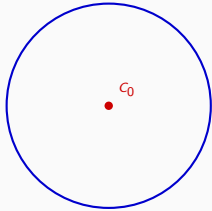
Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw the Generating Circle ($D = 50 \text{ mm}$, $R = 25 \text{ mm}$)

C_0 → Initial Center Point
Circle → Generating Circle

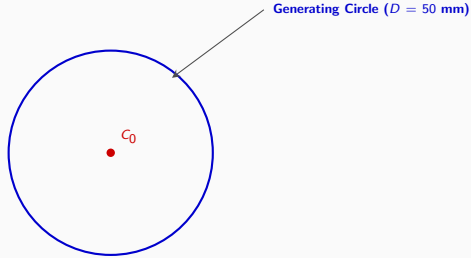


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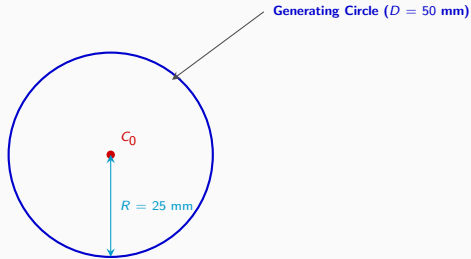
$C_0 \rightarrow$ Initial Center Point
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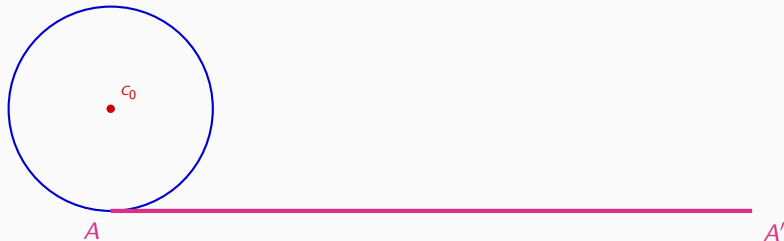
Step 1: Draw the Generating Circle ($D = 50 \text{ mm}$, $R = 25 \text{ mm}$)



$C_0 \rightarrow$ Initial Center Point
Circle $\rightarrow$ Generating Circle

Step 2: Draw Directing Line AA' ($\pi D = 157$ mm) and Axis Line

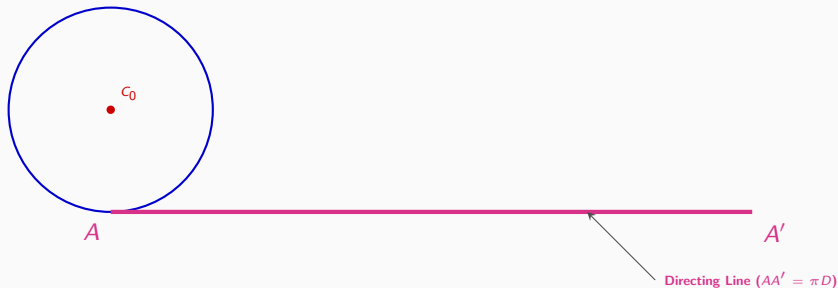
AA' → Directing Line (Base Line)
Axis Line → Center Line



Step 2: Draw Directing Line AA' ($\pi D = 157 \text{ mm}$) and Axis Line

AA' → Directing Line (Base Line)

Axis Line → Center Line



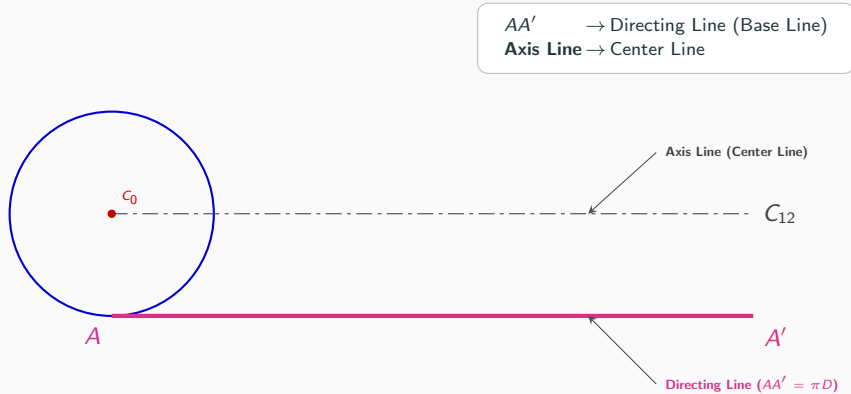
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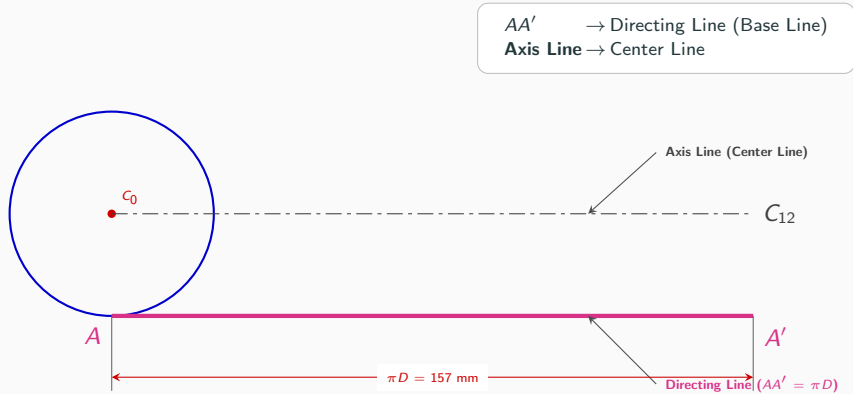
Axis Line → Center Line



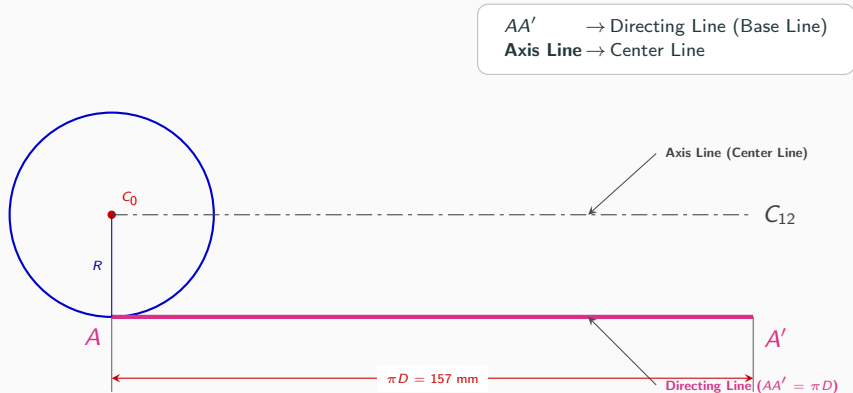
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Step 3: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
1 . . . 12 → Circle Divisions

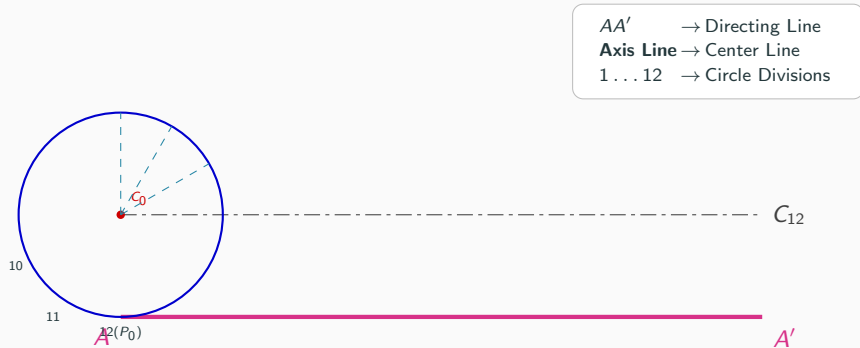


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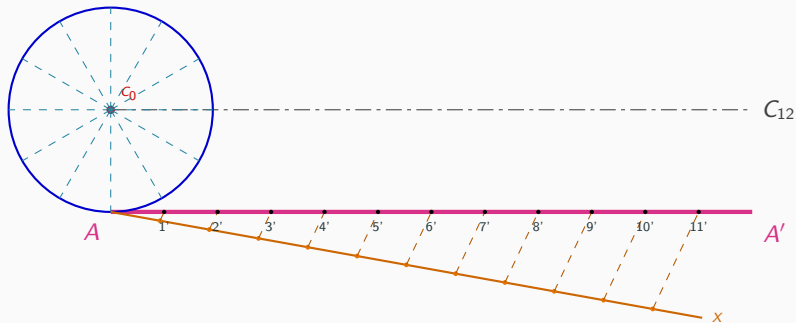


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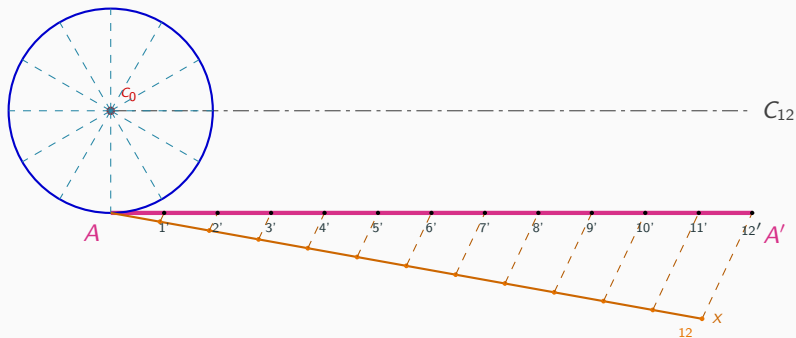
Step 4: Divide Directing Line AA' using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



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Step 5: Project Divisions to Locate Centers $C_1 \dots C_{12}$

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Axis Line → Center Line

$C_1 \dots C_{12}$ → Center Points



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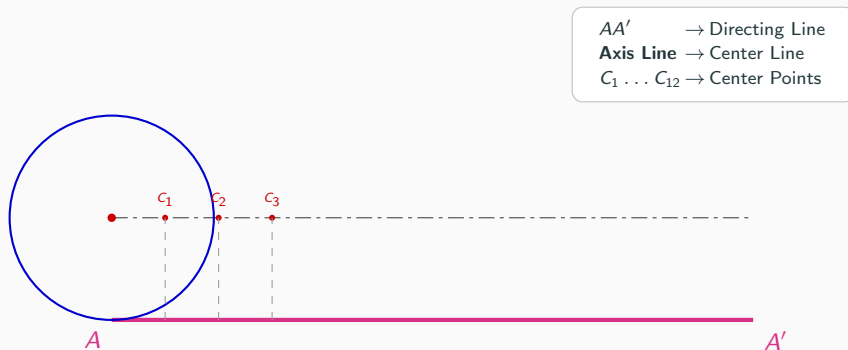
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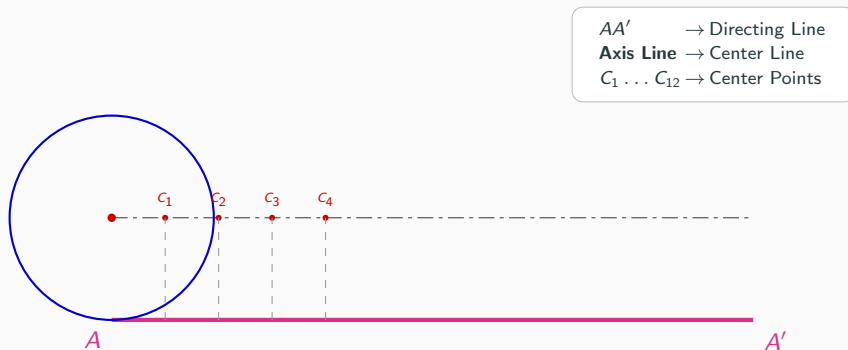
$C_1 \dots C_{12}$ → Center Points



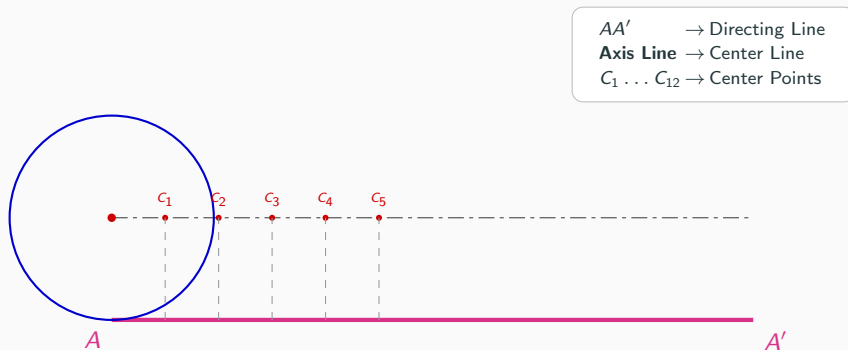
Step 5: Project Divisions to Locate Centers $C_1 \dots C_{12}$



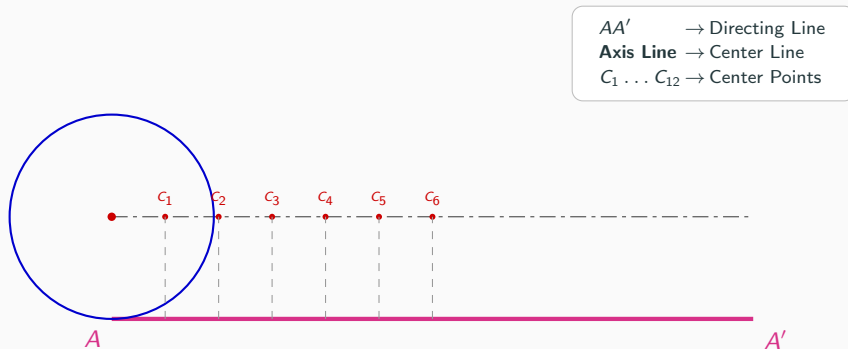
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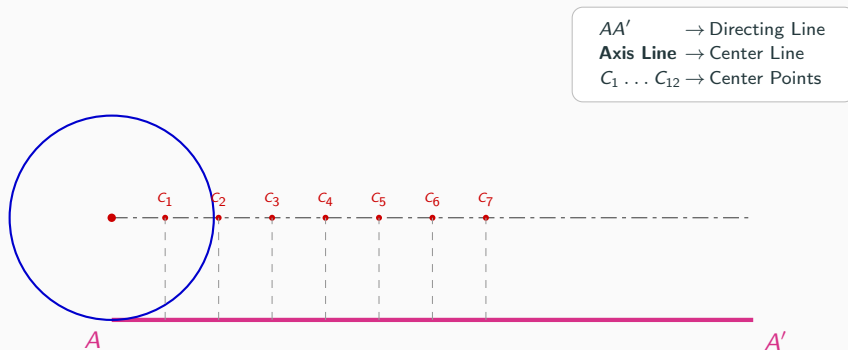
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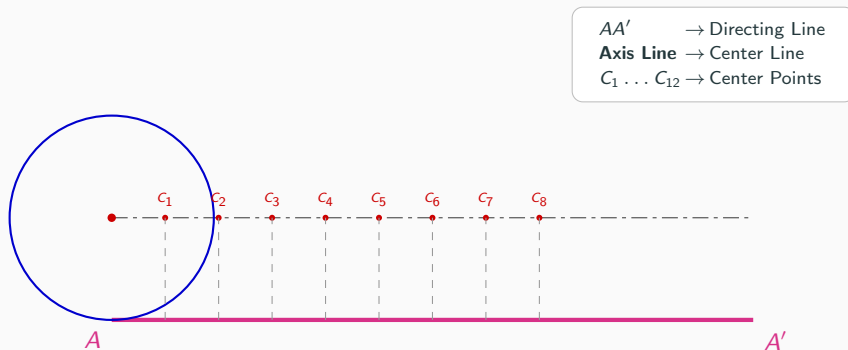
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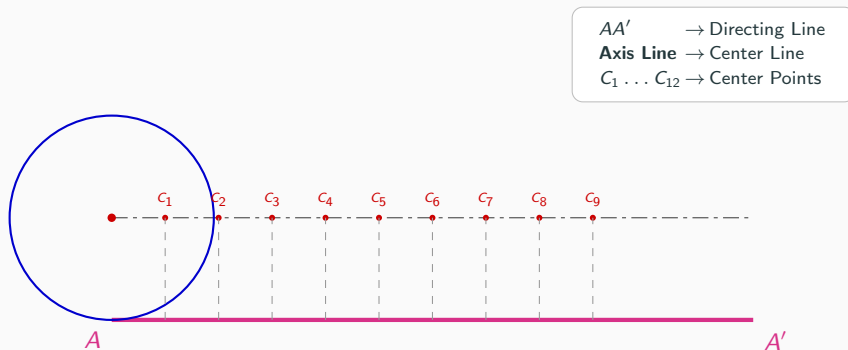
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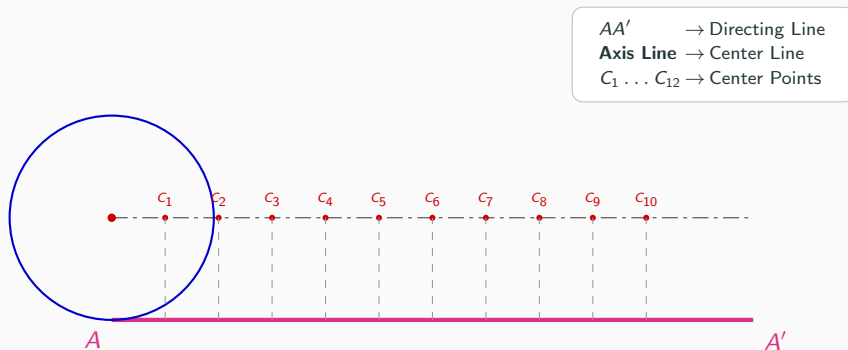
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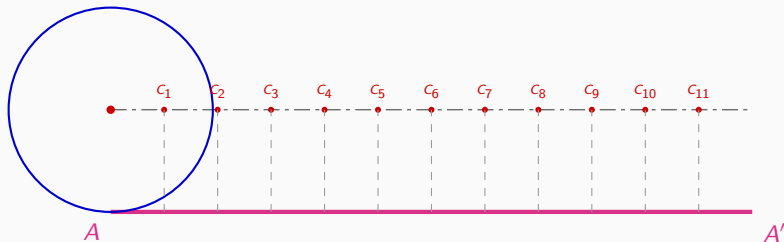


Step 5: Project Divisions to Locate Centers $C_1 \dots C_{12}$

AA' → Directing Line

Axis Line → Center Line

$C_1 \dots C_{12}$ → Center Points

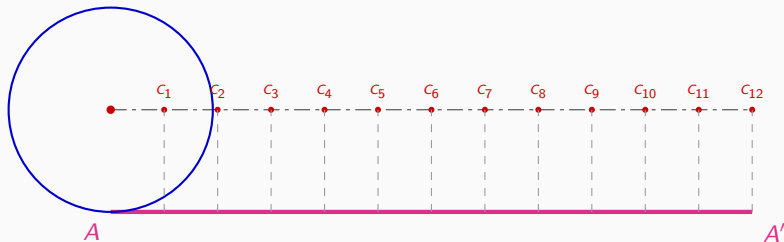


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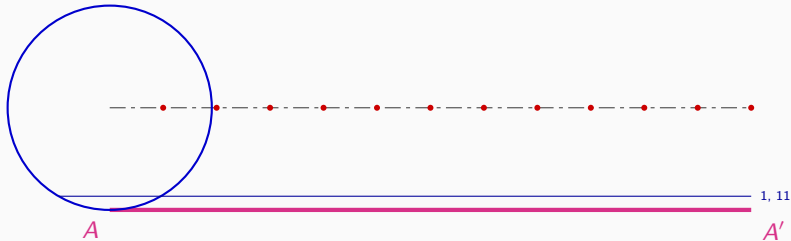
Axis Line → Center Line

$C_1 \dots C_{12}$ → Center Points



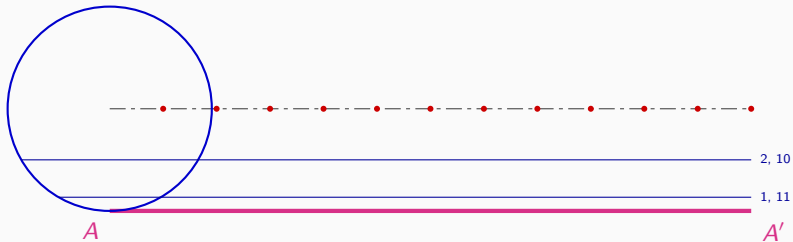
Step 6: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers



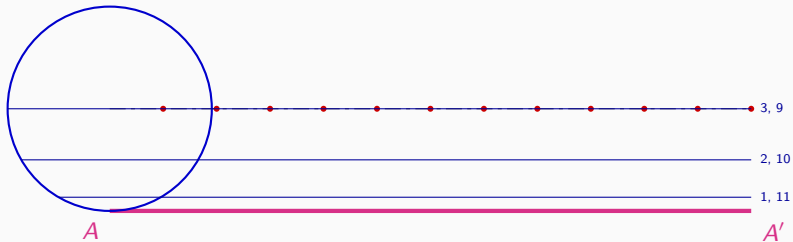
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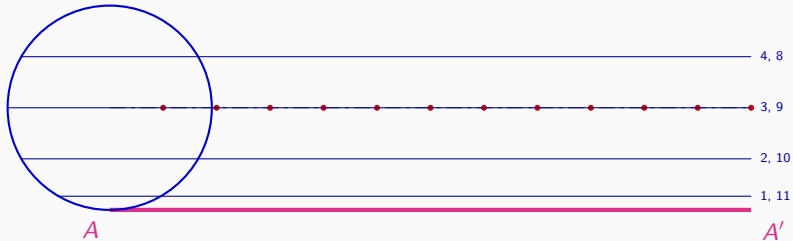
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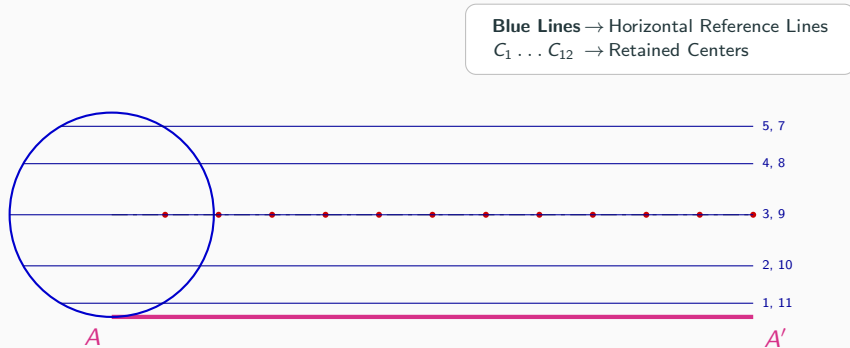


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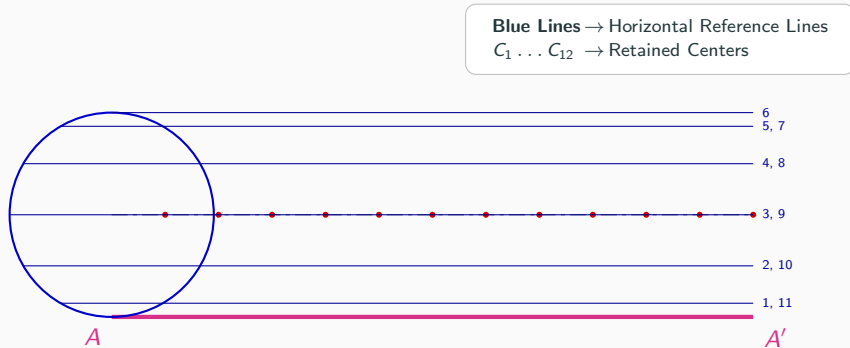
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Step 7: Strike Arcs ($R = 25$ mm) to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cycloid Locus Points



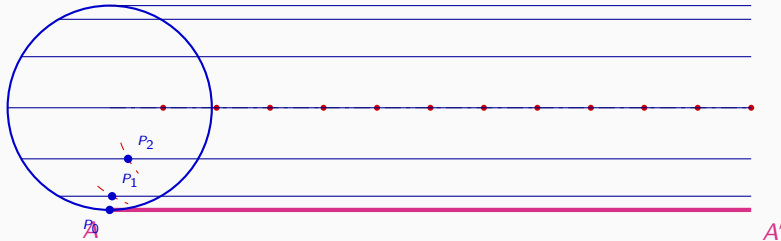
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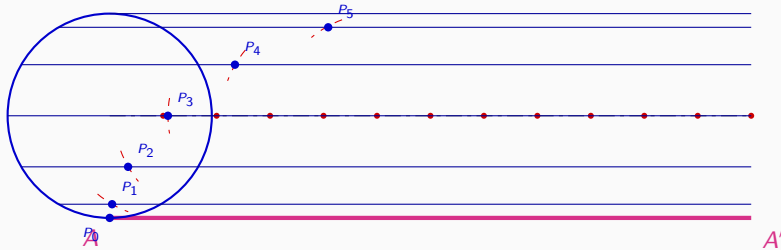
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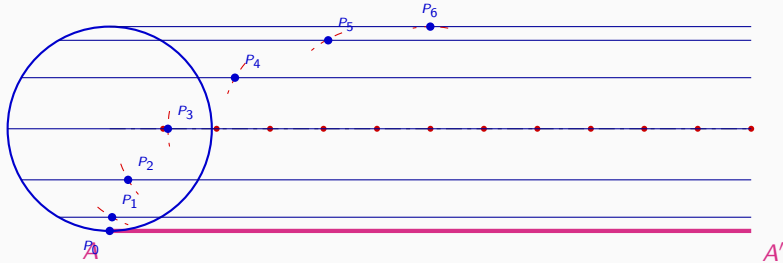


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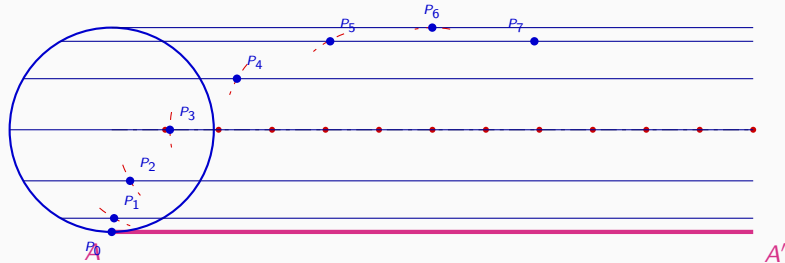
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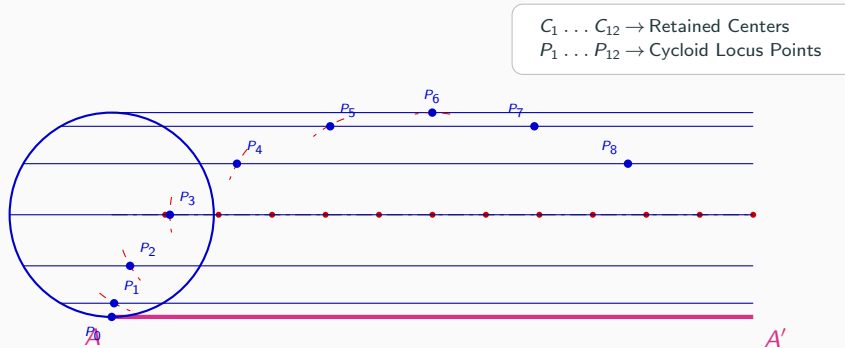
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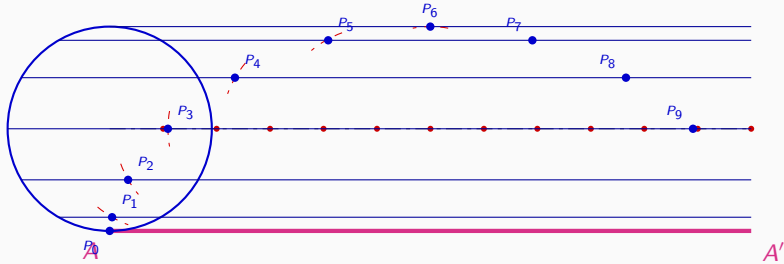


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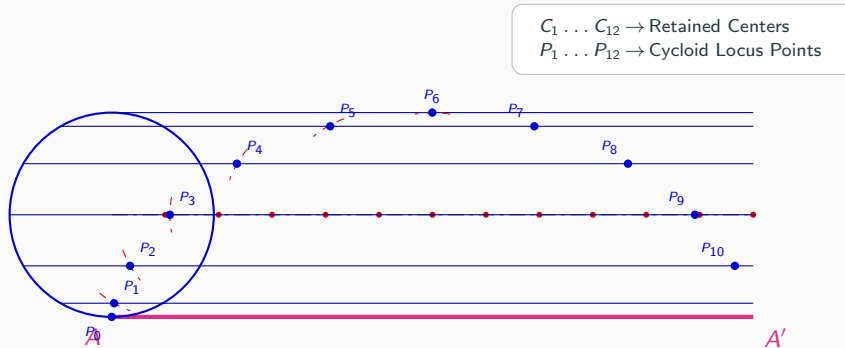
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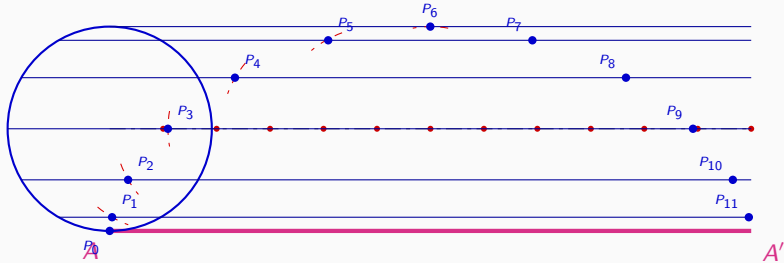
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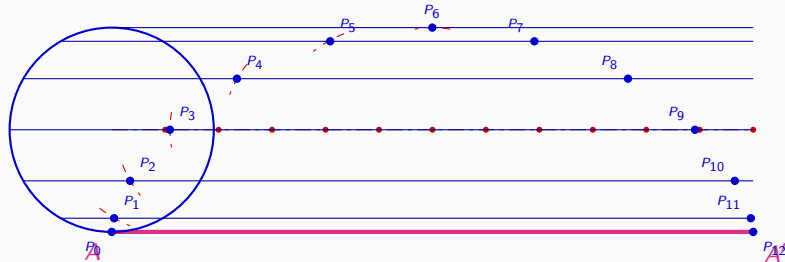
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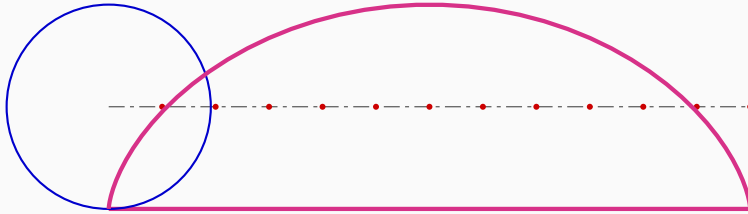
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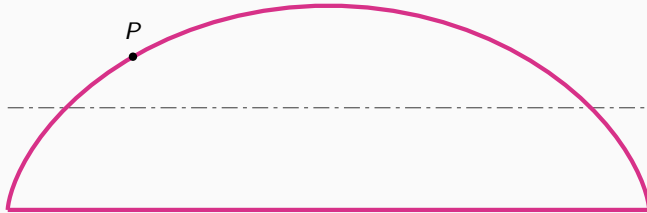
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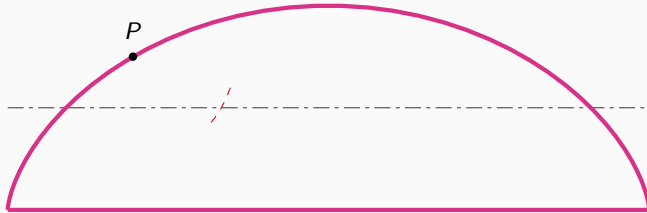
Step 8: Smoothly Join Points $P_0 \dots P_{12}$



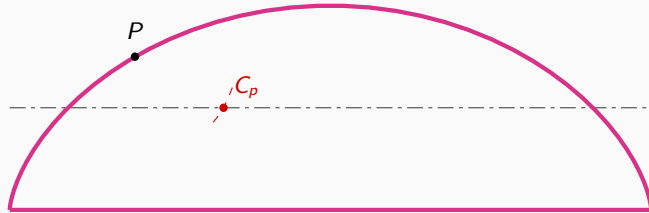
Step 9: Mark Point P & Strike Arc to Axis Line



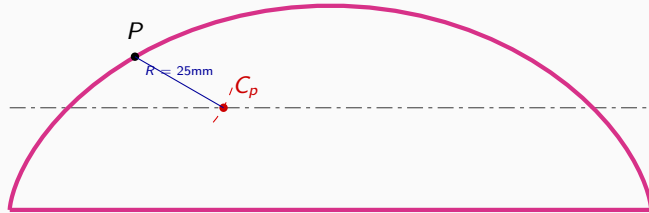
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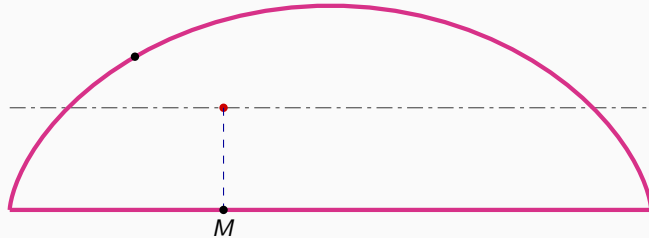
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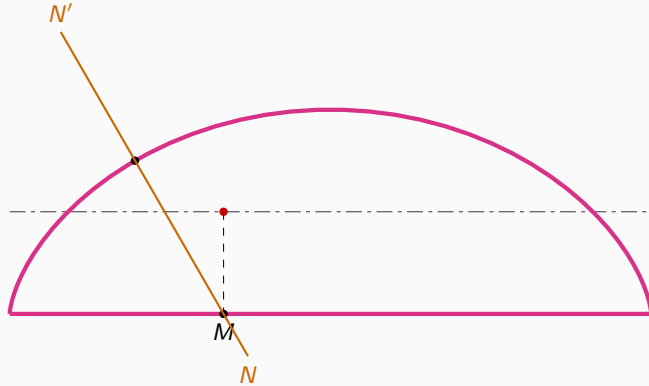
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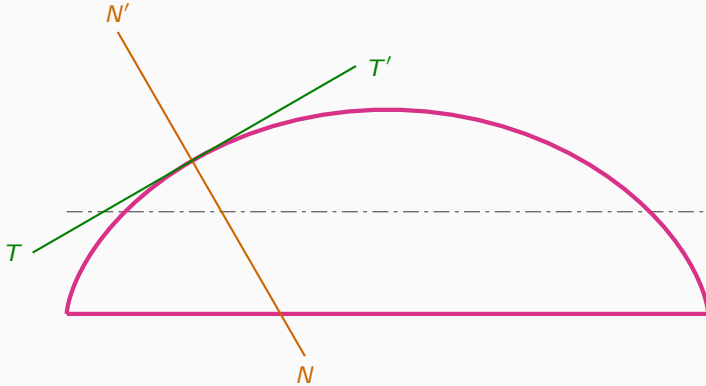
Step 10: Project C_p Vertically to Locate Normal $N-N'$



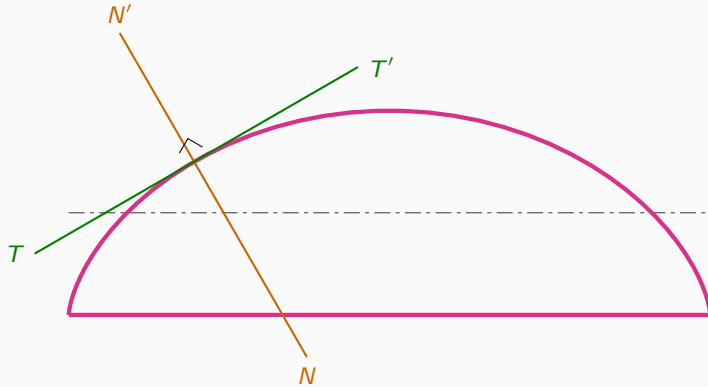
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Step 11: Draw Tangent Line $T-T'$ Perpendicular to Normal



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Step 12: Final Completed Cycloid with Tangent & Normal

